



Nebraska Public Power District

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10 CFR 50.73

NLS2025005
February 20, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2024-003-01
Cooper Nuclear Station, Docket No. 50-298, Renewed License No. DPR-46

The purpose of this correspondence is to forward Licensee Event Report 2024-003-01.

This letter does not contain regulatory commitments.

Sincerely,

Khalil Dia
Site Vice President

/bk

Attachment: Licensee Event Report 2024-003-01

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

NRC FORM 366 (04-02-2024)			U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.			EXPIRES: 04/30/2027		
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. Facility Name Cooper Nuclear Station						☒ 050 ☐ 052		2. Docket Number 298		3. Page 1 of 5	
4. Title High Pressure Coolant Injection Steam Leak Causes Condition that Could Have Prevented Fulfillment of a Safety Function and a Condition Prohibited by Technical Specifications											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
01	31	2024	2024	003	01	02	20	2025	Facility Name	☐ 052	Docket Number
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager								Phone Number (Include area code) (402) 825-5416			
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS		
E	BJ	TRP	Y010	Y							
14. Supplemental Report Expected								15. Expected Submission Date			
☒ No				☐ Yes (If yes, complete 15. Expected Submission Date)				Month	Day	Year	
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines) On January 31, 2024, a steam leak was identified near the High Pressure Coolant Injection (HPCI) drain line steam trap. This issue was entered into the work management process. The steam leak was later noted as degraded on February 19, 2024, and during a walkdown on February 22, 2024, to support the work order for repair, a maintenance technician identified steam potentially coming from the piping between the HPCI steam trap and a HPCI downstream valve. As a result of this potential through-wall pipe leak, Operations declared HPCI inoperable at 1718 hours on February 22, 2024, and entered the applicable Technical Specifications Condition. An eight-hour non-emergency event notification (EN 56989) was made to the Nuclear Regulatory Commission. The HPCI trap assembly (steam trap, outlet valve, and piping) was replaced and HPCI was returned to operable status at 2101 on February 24, 2024. A failure analysis on the removed HPCI steam trap assembly revealed internal damage resulting in leak-by, allowing flow accelerated corrosion (FAC) and/or erosion of the piping immediately downstream. The root cause is steam trap performance and leak-by events were not thoroughly understood regarding the potential adverse effects of erosion and/or FAC.											



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Cooper Nuclear Station	2. DOCKET NUMBER 05000-298	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
		2024	003	01

NARRATIVE

PLANT STATUS

At the time of the event on January 31, 2024, Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 100 percent steady-state power.

BACKGROUND

The High Pressure Coolant Injection (HPCI) system [EIS:BJ] provides protection to the core in case of a small break in the reactor coolant pressure boundary which does not result in a rapid depressurization of the reactor vessel [EIS:RPV]. The HPCI system permits the nuclear plant to be shutdown while maintaining sufficient reactor vessel water inventory until the reactor vessel is depressurized. HPCI continues to operate until reactor vessel pressure is below the pressure at which Low Pressure Coolant Injection system [EIS:BO] operation or Core Spray system [EIS:BM] operation can be used to maintain core cooling.

HPCI consists of a steam turbine assembly [EIS:TRB] driving a multi-stage booster and main pump assembly [EIS:P] and system piping valves, controls and instrumentation. The HPCI turbine is driven by steam from the reactor which is generated by decay and residual heat.

The HPCI turbine drain line steam trap, HPCI-TP-S57, provides a drain path for the HPCI steam supply drip leg since the HPCI steam supply block valve is normally closed preventing steam from supplying the turbine. Condensate forms from the steam in the drip leg. HPCI-TP-S57 allows the high pressure condensate to pass through it to the lower pressure drain line to the Main Condenser [EIS:SG]. The trap is designed to prevent high pressure steam from passing through the drain with the condensate. HPCI-TP-S57 is manufactured by Yarway Corporation, model 515BSWR.

HPCI is a single train system and is required to be operable to satisfy the requirements of Technical Specifications (TS) 3.5.1.

EVENT DESCRIPTION

On January 31, 2024, a station operator identified a steam leak near HPCI-TP-S57. The steam leak could be heard once entering the HPCI room and an approximate six-inch steam plume was visible around HPCI-TP-S57. This issue was entered into the work management process for repair. On February 19, 2024, during a subsequent station operator tour, the steam leak had degraded. Condensation was no longer completely contained in the catch basin; floor troughs and drains had completely captured the remaining condensation. The operability reviews for these conditions assessed the steam leak as coming from the body to bonnet gasket of HPCI-TP-S57 which is not considered American Society of Mechanical Engineers (ASME) code leakage (leakage from mechanical connections is classified as non-code leakage). Based on the leakage being non-code leakage, not impacting surrounding HPCI equipment, and HPCI steam pressure unaffected, HPCI remained operable.



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2024	003	01								

On February 22, 2024, during a walkdown performed prior to scheduled corrective maintenance for the HPCI steam leak, a maintenance technician discovered that the steam leak was potentially coming from drain line piping between HPCI-TP-S57 and downstream valve HPCI-V-130, HPCI steam line drip-leg drain trap outlet valve. HPCI was declared inoperable at 1718 hours on February 22, 2024, and Operations entered TS 3.5.1, Condition C.2 to restore HPCI to operable status within 14 days.

Following removal of piping insulation, it was determined the HPCI drain line piping had developed a through-wall leak located on the HPCI-TP-S57 downstream weld toe. The trap assembly (HPCI-TP-S57, HPCI-V-130, and drain line piping) was replaced and post-work testing completed. HPCI was declared operable at 2101 hours on February 24, 2024, and Operations exited TS 3.5.1, Condition C.

The removed piping was visually inspected by borescope and an ultrasonic test (UT) performed. The UT indicated significant loss of pipe thickness near HPCI-TP-S57. The HPCI drain line piping between HPCI-TP-S57 and HPCI-V-130 is one-inch, Class II carbon steel and is within the ASME Section XI Class 2 (high energy piping) boundary.

A failure analysis was conducted on the HPCI-TP-S57 and downstream piping. The analysis indicated the through-wall leak occurred due to two-phase flow accelerated corrosion (FAC). Assessment of the steam trap revealed internal damage resulting in leak-by and the excessive flow of wet steam through the trap outlet, resulting in liquid droplet impingement and erosive damage, catalyzing FAC of the downstream pipe. To this effect, the leak-by of the steam trap accelerated the rate of FAC degradation. A failure analysis was also performed on HPCI-V-130 which identified severe degradation of the plug/disk, seat, valve body, and attached piping due to FAC and erosion. HPCI-V-130 is susceptible to FAC and erosion due to the operating conditions of the system, adjacency to HPCI-TP-S57, and its construction of non-FAC resistant material.

Maintenance history of the HPCI-TP-S57 was also reviewed and revealed that in 1999, the preventive maintenance plan performance frequency was extended and subsequent steam trap leak-by occurrences identified in 2005 and 2012 were considered a station thermal performance issue only.

BASIS FOR REPORT

Since HPCI is a single-train system, this event is being reported per 10 CFR 50.73(a)(2)(v)(D), "Any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident."

An eight-hour non-emergency event notification (EN 56989) was also made to the Nuclear Regulatory Commission (NRC).

An engineering review was completed and concluded there was evidence HPCI was inoperable back to the date the steam leak was discovered; i.e., on January 31, 2024. Based on this information, HPCI was inoperable from January 31, 2024, until February 24, 2024. TS 3.5.1, Condition C.2, to restore HPCI system to operable status, applies which has a completion time of 14 days. Based on exceeding the TS completion



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time, this event is also reportable under 10 CFR 50.73(a)(2)(i)(B) as any operation or condition which was prohibited by TS.

SAFETY SIGNIFICANCE

Subsequent to declaring HPCI inoperable, Engineering evaluated the effects resulting from the HPCI steam line leak downstream of steam trap HPCI-TP-S57. The effects of the steam leak were determined for the identified through-wall hole size and conservatively assumed that HPCI-TP-S57 failed in the open position.

The results of this evaluation demonstrated that the temperatures and pressures remain acceptable for the equipment located in the affected spaces during the limiting conditions of a high energy line break and the post loss of coolant accident heat-up environment. The components in the affected areas are qualified for the maximum possible relative humidity and the environmental qualification program normal and accident total integrated doses remain bounding for the steam leak. The existing flooding analysis remains valid and bounding as the steam leak did not impact internal flood heights. Additionally, the steam diverted from the HPCI turbine by the steam leak does not impact HPCI system operation.

The evaluation concluded the HPCI system and components in the affected areas remained capable of performing their safety-related functions. Accordingly, this event will not count toward the NRC Safety System Functional Failure performance indicator.

The through-wall leak location was also evaluated for structural integrity. The analysis concluded the piping system downstream of HPCI-TP-S57 would have remained structurally intact during a seismic event under design conditions.

There were no consequences to general safety of the public, nuclear safety, or radiological safety.

CAUSE AND CORRECTIVE ACTIONS

The direct cause is steam trap leak-by allowed FAC and/or erosion of the piping immediately downstream of HPCI-TP-S57 resulting in a through-wall leak.

The root cause is steam trap performance and leak-by events were not thoroughly understood regarding potential adverse effects of erosion and/or FAC. The associated maintenance plan frequency for HPCI-TP-S57 was extended with the perception that steam trap performance affected station thermal efficiency only and without consideration for an increase in FAC/erosion of the downstream piping.

The following corrective actions have been taken and are complete:

- Replaced the HPCI trap assembly and drain line piping to correct the condition.
- Revised the maintenance plan frequency to refurbish HPCI-TP-S57 at a shorter frequency. This will



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reduce time in service to address degradation. Additionally, revised the maintenance plan to perform ultrasonic testing if evidence of leak-by is found to detect material loss.

- Added guidance to the associated maintenance procedure to initiate a condition report if indications of leak-by are found during refurbishment.
- Presented case study of the associated root cause evaluation to engineering support personnel and updated training lesson "Determine Preventive and Predictive Maintenance Requirements for Selected Components" to reference the root cause evaluation for the event. These actions reinforce the need to consider FAC/erosion aspects when initiating or reviewing maintenance plan change requests.
- Added a step to the preventive maintenance plan procedure for FAC program engineer review of maintenance plan changes if there is the potential to affect flow through FAC or erosion corrosion susceptible piping. This will prompt discussions with the FAC program engineer if the impact is unknown.
- Also added and revised maintenance plans for similar high pressure steam supply drain lines with steam traps.

PREVIOUS EVENTS

A review of CNS licensee event reports (LER) for the last three years was performed. There were no previous occurrences involving a similar failure to that discussed in this report. The following conditions related to HPCI were reported:

LER 2024-005-00 – HPCI pressure switch found outside of calibration tolerance due to not being installed per seismic test report mounting requirements [May 13, 2024].

LER 2022-003-00 – Inadvertent HPCI injection into the RPV during HPCI beginning of cycle surveillance testing. The injected water increased reactivity and caused a reactor scram [January 11, 2023].

LER 2021-003-00 – During surveillance testing, HPCI flow indicating switch, HPCI-FIS-78, failed to trip prior to exceeding the TS 3.3.5.1 (Emergency Core Cooling System Instrumentation) function for HPCI pump discharge flow-low (bypass) [December 27, 2021].